Xiahua Liu

Senior Software Engineer | C++ / Rust / Python / Linux / Robotics
Pleasanton, CA
Website: xiahua.pages.dev LinkedIn: linkedin.com/in/xiahualiu GitHub: github.com/xiahualiu

Professional Summary

- 6+ years of experience in biotech / aerospace / automotive / railroad systems.
- Hands-on experience working with cross-functional teams to bring commercial products to customers.
- Research background in robotics, control theory, machine learning (ML), and reinforcement learning (RL).
- Full system optimization at both the kernel and userspace levels to meet high throughput or low latency goals.

Technical Skills

Programming Languages: C++ (11/14/17/20), Rust, Python, MATLAB
Controls & Estimation: Control theory, extended Kalman filter (EKF), reinforcement learning (RL), etc.
Linux: Device drivers and kernel customization, upstream systemd contributor, Secure Boot, LUKS
Real-Time Operating Systems: FreeRTOS, QNX, RT-Linux
Robotics Software: Robot Operating System (ROS), MATLAB/Simulink
Image Processing: OpenCV, autofocus algorithms, feature extraction, imaging system calibration
Systems: Real-time concurrent programming, gRPC / REST API design, state machines
Networking: Routing / VPN / NAT and firewall configuration, network troubleshooting
Processors & Microcontrollers: ARM, RISC-V, x86, ESP32, NXP S32K144
Automotive Standards: ISO 26262, SAE J1939, SAE J1979, ISO 15765-4

Work Experience

Software Engineer III

10x Genomics | May 2025 - Oct 2026 | Pleasanton, CA

- **Led the development of Secure Boot and LUKS OS security features for Atera®.**
  - Managed kernel and device driver releases and signing on AWS CloudHSM and KMS.
  - Developed QEMU end-to-end (E2E) boot tests with Open Virtual Machine Firmware (OVMF).
  - Supported OTA upgrades of both the UEFI UKI image and Linux drivers.
- **Collaborated with the Teledyne DALSA® Europe team on Linux device driver development.**
  - Adjusted kernel variables to optimize the TLB cache for maximum throughput.
  - Enabled kernel features for framegrabber driver troubleshooting.
- **Developed Python/C++ optical calibration software.**
  - Developed a path planning algorithm to reduce calibration time from 6–8 hours to 30 minutes.
  - Calibrations include chromatic aberration, point spread function (PSF), field flatness, distortion, etc.
  - Achieved subpixel (< 100 nm) XYZ alignment precision using a customized PCA algorithm.
- **Developed hardware and software autofocus algorithms using OpenCV (C++ and Python).**
- **Collaborated with optics, electrical engineering (EE), camera, manufacturing, and field service teams to troubleshoot issues and support Atera's transition from the lab into the field.**

Senior Software Engineer

Wabtec | May 2023 - May 2025 | Melbourne, FL

- **Maintained the large-scale production C++ codebase for Trip Optimizer®.**
  - Migrated legacy C++98 code to C++11 and fixed various low-level race conditions.
  - Compared data operations with those of modern data systems to highlight non-ACID operations.
- **Improved software testing processes for the production codebase.**
  - Developed a VS Code Dev Container for software development and testing.

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Work Experience (continued)

Lead Control Engineer
GE Aerospace / Hyliion | Dec 2021 - Jan 2023 | West Chester, OH

- **Developed embedded C++ control software for KARNO® using MATLAB and Simulink.**
  - Designed circuits and PCBs for Type K thermocouples, flame rectification, and emergency stops, etc.
  - State machines for cold/warm start sequences and combustion temperature regulation.
- **Collaborated with manufacturing teams to troubleshoot issues and support the product's transition toward deployment.**

Software Engineer
Clean Earth Rovers LLC | Jan 2021 - Nov 2021 | Cincinnati, OH

- **Led C++/ROS software development for Piranha®, the company's first water-surface robot prototype.**
  - Selected controllers ranging from PX4 to Raspberry Pi and ESP32.
- **Developed nonlinear control algorithms using feedback linearization for the robotic prototype.**
  - Modeled robot dynamics using SolidWorks and MATLAB.
- **Implemented an extended Kalman filter (EKF) for robot state estimation.**
  - Referenced open-source EKF code from PX4 and ArduPilot.
  - Wrote the EKF model in pure Python.

Robotics Researcher
IRAS, University of Cincinnati | Aug 2020 - Dec 2021 | Cincinnati, OH

- **Researched control theory and reinforcement learning for robotic manipulation at IRAS.**
- **Used OpenCV for computer vision and image processing in robotics research.**

Embedded System Engineer Intern
Blueway Electric Appliances Co., Ltd. | Sep 2017 - Jun 2018 | Guangdong, China

- **Implemented OBD-II for a battery management system using SAE J1979 and ISO 15765-4.**
- **Deployed battery-management diagnostics on the NXP S32K144 SoC running FreeRTOS.**

Open Source Projects

PawnDB | Real-Time Database
Designed and developed a lightweight database management system for real-time systems, implementing two-phase locking (2PL) for concurrency control.

Upstream contributions
systemd, Microsoft WSL, Visual Studio Code, Espressif ESP-IDF
Owner of a GitHub project with 100+ stars.

Education

Master of Science in Aerospace Engineering
University of Cincinnati | 2019 - 2021
GPA: 3.8/4.0; full graduate scholarship
Thesis: vehicle dynamics simulation, trajectory tracking, nonlinear control, and state estimation.
github.com/xiahualiu/ms-thesis

Bachelor of Science in Automotive Engineering
Beijing Institute of Technology | 2014 - 2018
Contributor to the official graduate thesis LaTeX template on Overleaf